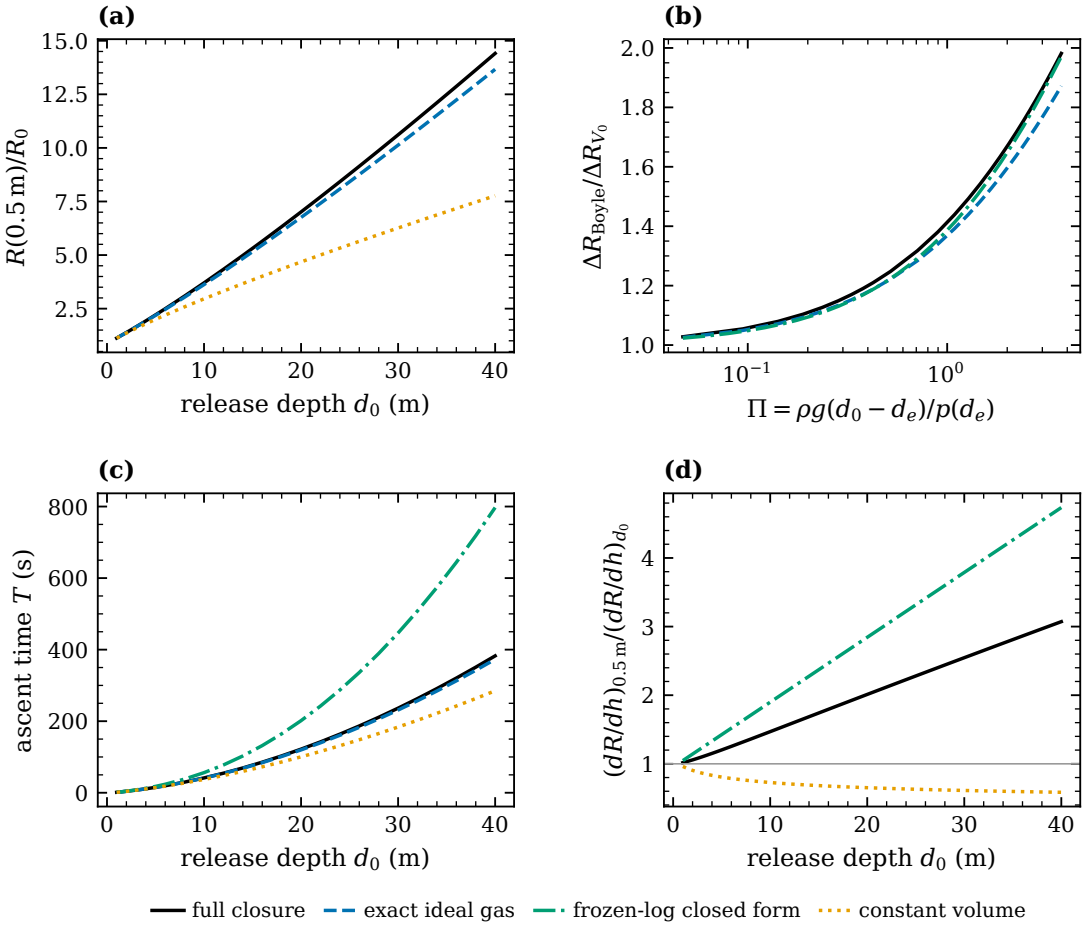
(a)
15.0
12.5
10.0
7.5
5.0
2.5
$R(0.5\,\mathrm{m})/R_0$
0
10
20
30
40
release depth d_0 (m)
(b)
2.0
1.8
1.6
1.4
1.2
1.0
$\Delta R_{\mathrm{Boyle}}/\Delta R_{V_0}$
10^{-1}
10^{0}
$\Pi = \rho g(d_0 - d_e)/p(d_e)$
(c)
800
600
400
200
0
ascent time T (s)
0
10
20
30
40
release depth d_0 (m)
(d)
4
3
2
1
$(dR/dh)_{0.5\,\mathrm{m}}/(dR/dh)_{d_0}$
0
10
20
30
40
release depth d_0 (m)
full closure
exact ideal gas
frozen-log closed form
constant volume